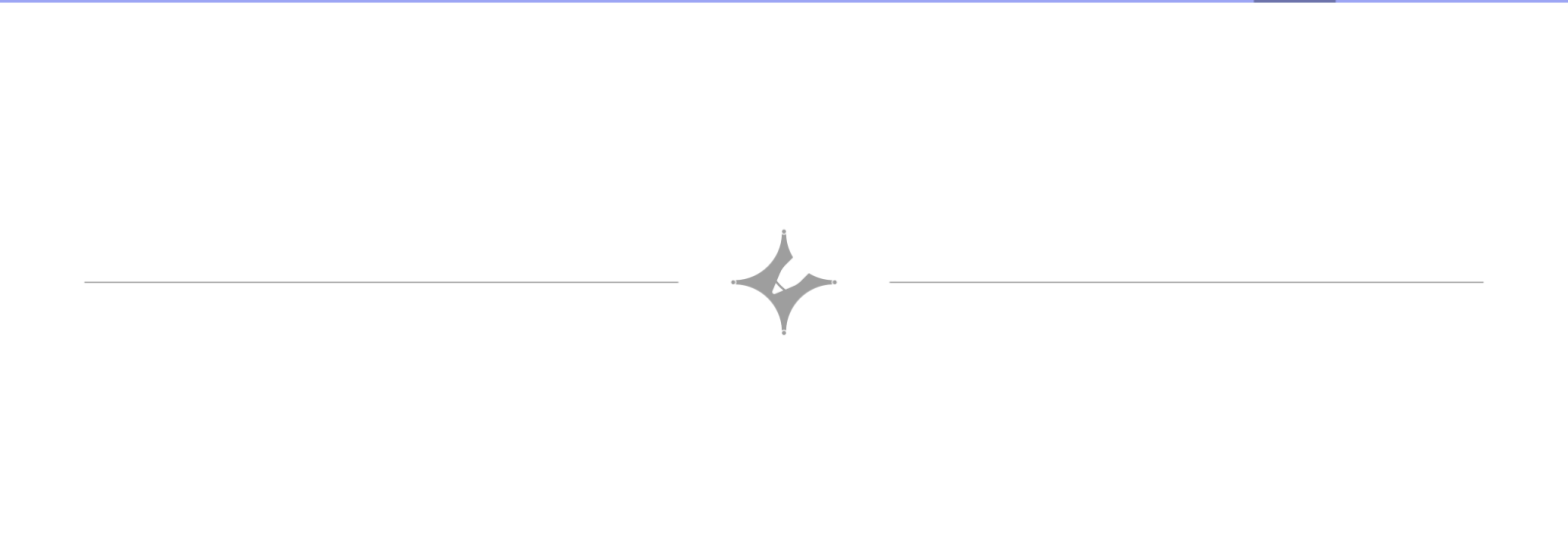
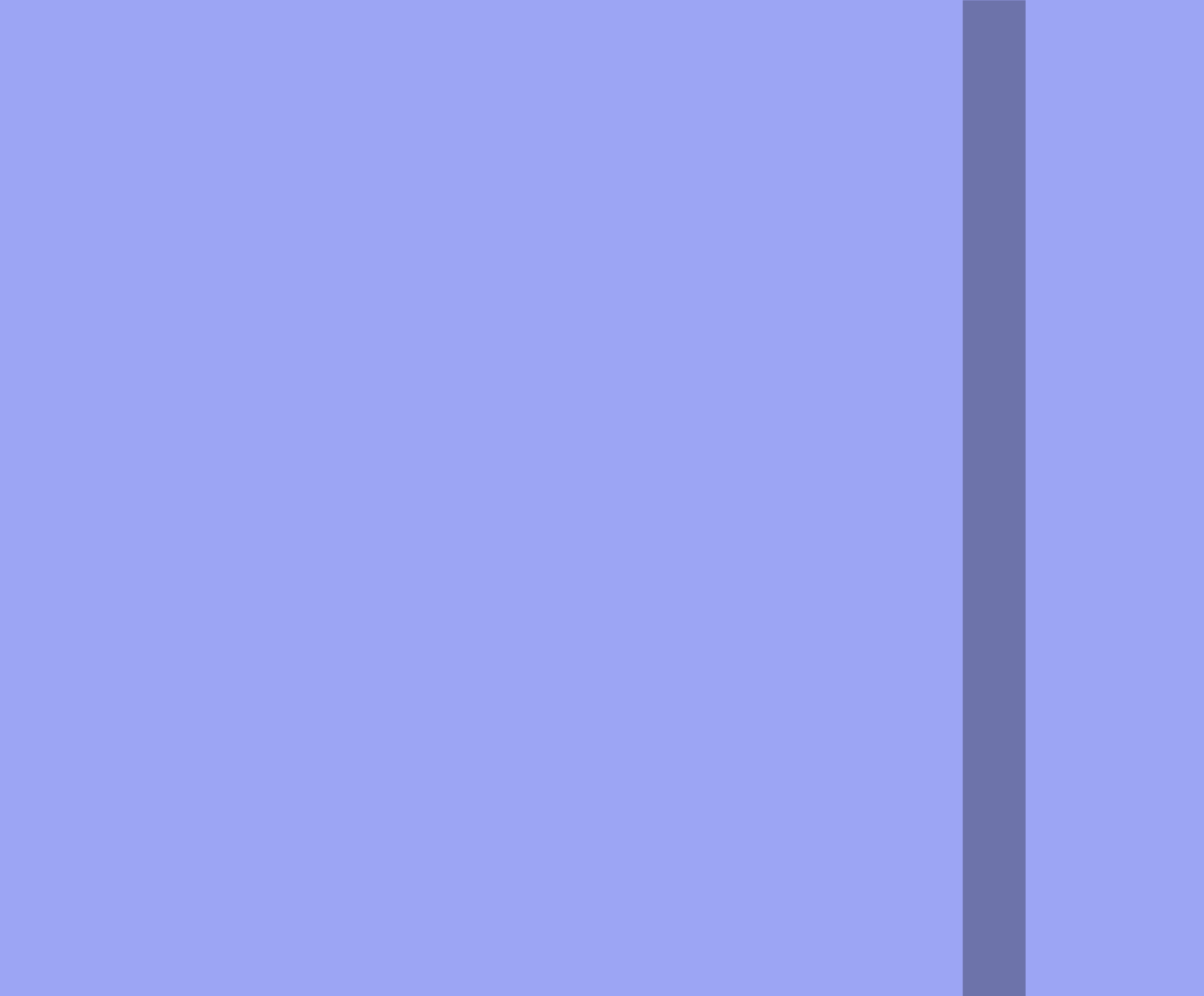




Chapter 10 随机化算法

定义: RNAND-TM program
 $= \text{NAND-TM} + \{ = \text{RAND}() \}$

Probabilist TM

$$\delta(P, a) = \begin{cases} (q_1, b_1, D_1) \\ (q_2, b_2, D_2) \end{cases} \quad \text{with equal probability}$$

定义: BPP.

F in BPP if $\exists$ RNAND-TM P

s.t. $\forall x \in \{0, 1\}^n$

$$\begin{cases} P \text{ halts within } a|x|^b \text{ step} \\ P_r(P(x) \neq F(x)) \leq \frac{2}{3} \end{cases}$$

Amplification Lemma.

Let $F: \{0,1\}^* \rightarrow \{0,1\}$, if $\exists$ poly-time

RAND-TM P , s.t. $\forall x \in \{0,1\}^*$

$$\Pr(P(x) = F(x)) \geq 1 - \delta, \quad \delta < \frac{1}{2}.$$

1) $\exists$ poly-time RAND-TM Q , s.t. $\forall x \in \{0,1\}^*$

$$\Pr(Q(x) = F(x)) \geq 1 - 2^{-p(x)}.$$

Q : on input x

1. run P on x for 2^k times

2. return the majority.

$$\Pr(Q \text{ fail}) = \sum_{i=k}^{2k} \binom{2k}{i} \delta^i (1-\delta)^{2k-i}$$

$$\leq \delta^k (1-\delta)^k \sum_{i=k}^{2k} \binom{2k}{i}$$

$$\leq \delta^k (1-\delta)^k 2^{2k}.$$

$$= (4\delta(1-\delta))^k$$

$$k = \frac{-p(n)}{\log 4\delta(1-\delta)}$$

$$= 2^{-P(n)}.$$

证:

$$F \in BPP \Leftrightarrow \exists G \in P, \text{ s.t. } \forall x \in \{0,1\}^*$$

$$\Pr[G(xr) = F(x)] \geq \frac{2}{3}$$

$$r \sim \{0,1\}^{a|x|^b}$$

$$P \subseteq BPP \subseteq EXP$$

证:

$$BPP \subseteq P/poly$$

ii: $\forall F \in BPP \Rightarrow \exists \text{ poly-time RNAND } P$

$$\Rightarrow \exists \text{ poly-time NAND-TM } Q.$$

$$\Pr_{r \sim \{0,1\}^{a|x|^b}}[Q(xr) = F(x)] \geq \frac{2}{3} \quad (\text{放大到 } 1 - \frac{0.1}{2^n})$$

$$\Rightarrow F(x) \text{ } |x|=n$$

$$\sum_{|x|=n} \Pr[Q(xr) \neq F(x)] \leq \frac{0.1}{2^n} 2^n \leq 0.1$$

$$\therefore \Pr(Q(x,r) = F(x) \mid x \text{ with length } n) \geq 1 - 0.1 = 0.9.$$

$\Downarrow$.

$$\exists r \text{ with } |r| = an^b \text{ s.t.}$$

$$Q(x,r) = F(x) \text{ for all } x \text{ with length } n.$$

#.

Pseudo random Generator.

A function $G : \{0,1\}^l \rightarrow \{0,1\}^m$ is a (T, ϵ) -pseudorandom generator, if $\forall$ circuit C with m inputs, at most T gates.

$$\left| \Pr_{s \sim \{0,1\}^l} [C(G(s)) = 1] - \Pr_{r \sim \{0,1\}^m} [C(r) = 1] \right| \leq \epsilon.$$

$$\text{Optimal : } m = 2^l, \epsilon = 2^{-l}$$

Lemma.

If $P = NP$, then $BPP = P$

